Chapter 18 worked solutions • The Linux system

OS18-E01 • Fork and exec

1. The child has a separate task and mm, with required page-table structures. This resident private anonymous page is initially shared with write protection. Under the stated no-write interval, reading copies no page content; exec then replaces the old mapping. Task, mm, and page-table setup still costs work. A write fault copies when separation is needed, but a safely exclusive page can be reused. This example does not cover MAP_SHARED or all VMAs.
2. In this single-thread case, child PID persists while code, data, stack, entry are replaced. Parent remains independent. Multithreaded exec has additional thread termination rules, not another ordinary fork.
3. Early failure returns to old code; record errno and explicitly _exit as appropriate so parent wait observes failure. After v3.2 ELF flush_old_exec, some failures deliver fatal signals without restorable old image. Parent distinguishes ordinary exit from signal termination. Identify task/address space at every step.

OS18-E02 • Module references

1. No, under the stated lifetime protocol and non-forced unload: the handle and in-flight request each retain a reference and may still call module code.
2. Block new legal entry/references, complete/cancel I/O with DMA stopped, close handles, and wait for actual worker/timer/RCU protocols. Free only when references and callbacks no longer use code. There is no universal one-RCU-wait rule.
3. No. Signatures support provenance/integrity policy, not pointer/DMA lifetime. Universal modern signature enforcement is not assumed for v3.2. References, I/O, and quiescence are separate conditions.

OS18-E03 • VFS and page cache

1. The call observed those cache bytes, not necessarily identical persistent media.
2. Dirty pages or block/controller caches may remain unwritten. Ordinary reads do not require durable writes.
3. A successful fsync of the specified file before power loss, with filesystem and device honoring flush contracts, requires its relevant dirty data and required metadata to reach the promised persistent layer. It does not make unrelated files or other unsynchronized state durable; fsync cannot be run after power loss. Visibility, writeback, and durability differ.

OS18-E04 • Credentials and regular-file DAC

1. In question order, read/write/execute are allow/deny/deny; allow/allow/deny; deny/deny/deny; allow/deny/deny. Ordinary DAC selects the caller's class before capability fallback. DAC_OVERRIDE cannot supply regular-file execute when every x bit is 0.
2. Mode 0401 makes S_IXUGO nonzero, permitting its execute DAC override. Noexec, ELF format, and other policies still apply. This is one regular-file DAC layer, not unrestricted root success.
3. No. IDs become 1000/1000/0; saved 0 permits restoration. Permanent dropping also addresses saved IDs, groups, capability sets, and privileged handles. UID values alone do not prove a sandbox. Derive from actual class/mode/capability branches.

OS18-E05 • Timeout, EIO, and late completion

1. No. Timeout means no observed completion by a deadline. If F40 is reassigned, late DMA corrupts its new owner. Obtain cancel/reset/other hardware-stop evidence before unmapping.
2. Under the stated single-terminal protocol, publish one notification with EIO and upper-layer transferred=0. This does not mean that DMA wrote no bytes to F40 before the error. Omitting error notification strands the waiter and reference cleanup.
3. Same-generation terminal duplicates cannot notify/free again. Old generation 3 cannot complete generation 4. Identity checking does not replace stopping DMA.
4. No. Consumer can still read/copyF40. Reuse requires DMA stopped, terminal processing completed, and all content references released. Notification and reclamation are different events;retain ownership, count, bytes, and reference state.

OS18-E06 • Clone sharing and ELF zero-fill

1. Counting parent+child, fork has 2 tasks, 2 mm, 2 fd tables, but both original fd4 entries retain O1. Thread-like clone has 2 tasks, 1 mm, 1 table, same O1, shared TGID. Neither shares the task_struct object itself.
2. Both fail EINVAL. THREAD requires SIGHAND;SIGHAND requires VM. Flags are not independent booleans, and shared VM alone does not imply sharedTGID.
3. Page 0x400000 contains 4096 file bytes; 0x401000 contains 2048 file + 2048 zero; 0x402000 contains 4096 zero. Totals 6144+6144=12288 virtual B. VMA states source/permissions; residency can wait for faults.
4. Validation failure before destroying the old image can return normally. After v3.2 flush_old_exec, restoration is not guaranteed; some errors terminate by SIGKILL/SIGSEGV. Successful single-thread exec preserves PID and replaces image rather than forking. Separate task/mm/fd/O1, virtual/resident, and commit boundaries.

OS18-E07 • IRQ reentry and unload races

1. Preemption disable blocks task rescheduling, not IRQs. Handler spins on L while its interrupted holder cannot release it.
2. Local masking does not stop CPU1. All participating contexts use an atomic spinlock, with irq-save/restore for local reentry. NMI/nonparticipants remain outside the guarantee.
3. In this specified blocking, non-forced model the counts are 5, 5, 2, 0, 0. GOING blocks entry. At zero, wait for actual worker, timer, RCU, DMA protocols to quiesce, then free. No universal single-RCU shortcut exists.
4. Another CPU can acquire a reference between zero-check and withdrawal, then execute freed code. State checking/reference acquisition need one lifetime protocol before draining. Prove local reentry, SMP exclusion, and last-reference races separately.

OS18-E08 • TCP segmentation and latency causes

1. With the stated effective MSS and packetization, 6000=4 • 1460+160 gives five payloads; headers are 5 • 40=200 B and L3 traffic is 6200 B. MTU and header sizes alone bound a segment but do not guarantee this segmentation. Ethernet, ACK, retransmissions, and offload change denominators.
2. No. The API accepted bytes. TCP acknowledgment concerns peer protocol state, not application consumption or remote durable storage.
3. Queue 2 ms, dispatch-to-transmit 1 ms, transmit-to-ACK 10 ms, total 13 ms. These linked timestamps support the decomposition; overlapping aggregate layer averages cannot be arbitrarily added.
4. No. Counters lack same-flow timing links. Collect request/flow timestamps, queue entry/exit, drop, ACK, retransmission evidence or controlled interventions to distinguish queueing, loss, reordering. State application/wire/ACK/completion denominators.

OS18-E09 • Pipe EOF, signals, and semop

1. Parent's retained writer prevents EOF. Empty buffer means blocking/EAGAIN until all writers close;only then does read return 0. Fork does not close unused endpoints automatically.
2. No. Repeated generation of the same blocked standard signal in the same pending queue can coalesce into one pending event. Realtime signals can queue/hold limited payload through corresponding APIs, with queue/handler constraints.
3. No. Atomic batching publishes no partial decrement while waiting. Nonblocking failure also cannot partially subtract. S0 cannot be permanently consumed first.
4. Consumer sees false;producer sets true/wakes before consumer enqueues;consumer then sleeps and loses the notification. Establish wakeable waiting state, then recheck;skip sleep if true and recheck after wake. Actual kernel primitives supply locking/ordering too. Stream data, writer lifetime, pending signals, and semaphore transactions are different state models.

References and attribution

Lin Cing, handwritten operating systems notes. Silberschatz et al., Operating System Concepts, 9th ed., Ch. 18. Linux v3.2 fork, memory, module, DAC, ELF. fork(2), signal(7).

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https://github.com/xinhi6716/computer-science-self-study